True bearings

1 State the true bearing of the following:

a Southwest

b Northwest

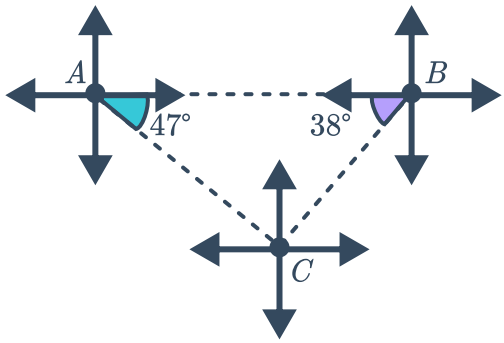
c West-Northwest

d South-Southeast

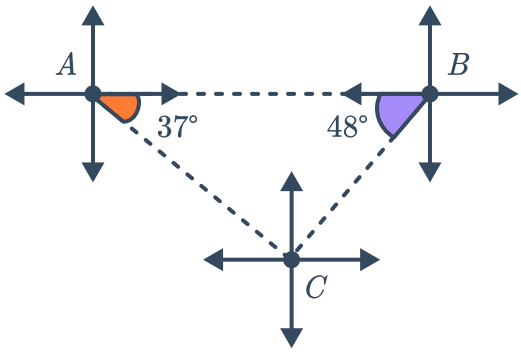
e Northeast

f Southeast

2 Determine the true bearing of point B from point A in the following bearings diagram:



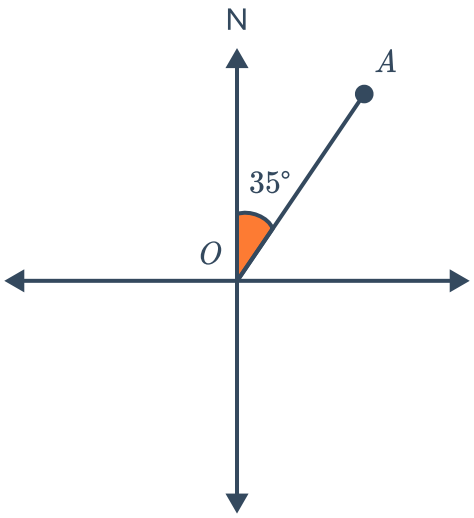
3 Determine the true bearing of point A from point B in the following bearings diagram:



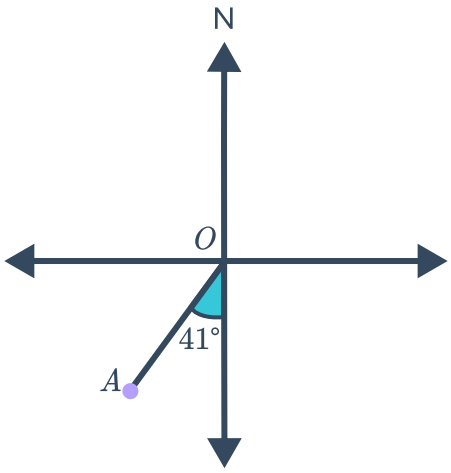
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4 Determine the true bearing of point A from O in the folowing diagrams:

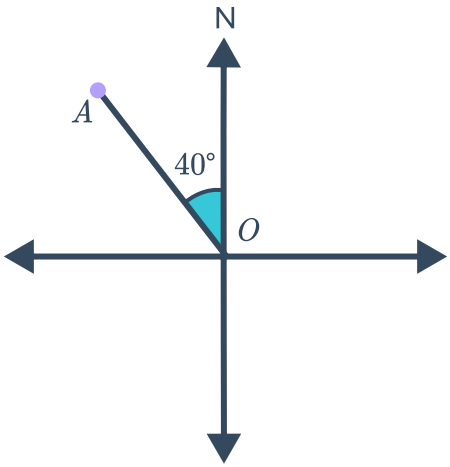
a



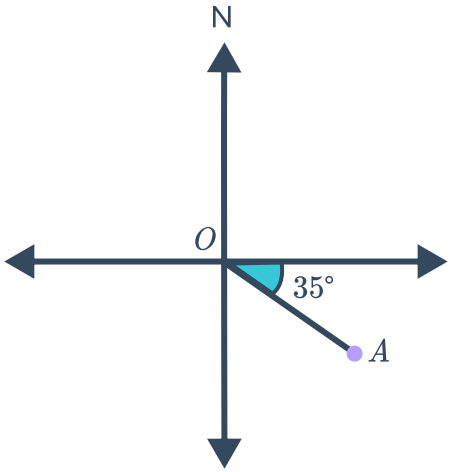
b



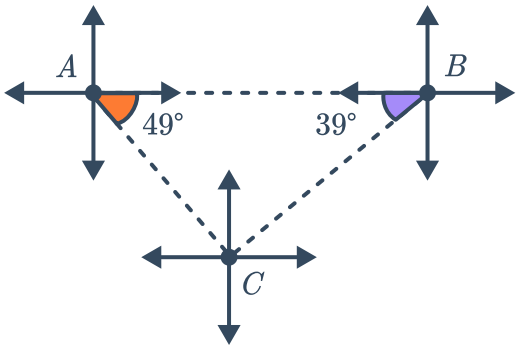
c



d

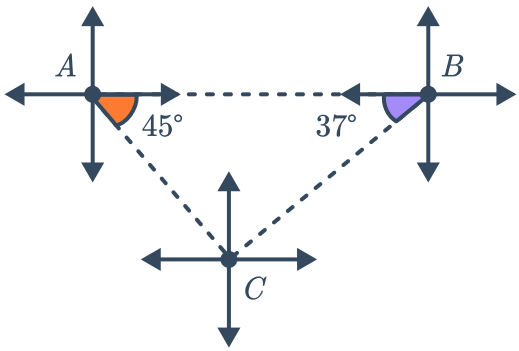


5 In the figure below, point B is due East of point A. Find the true bearing of point A from point C.

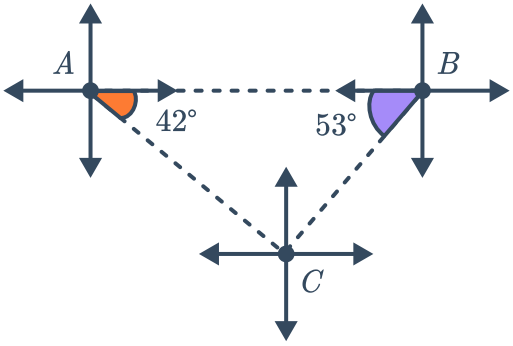


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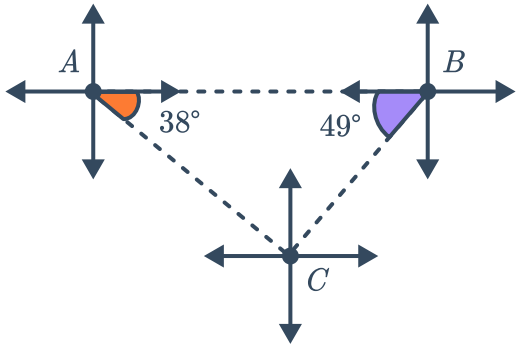
6 In the following diagram, find the true bearing of point B from point C.



7 In the following diagram, find the true bearing of point C from point A.



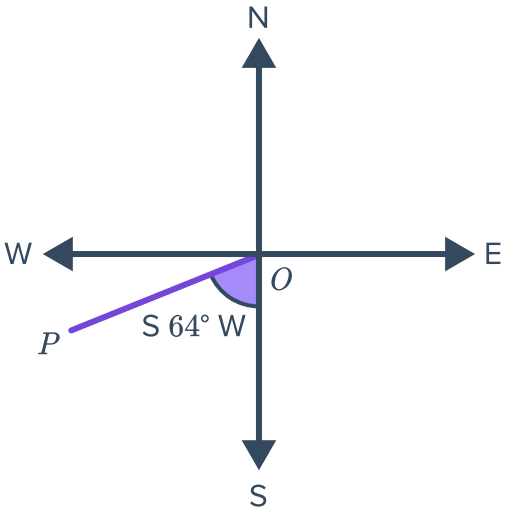
8 In the following diagram, find the true bearing of point C from point B.



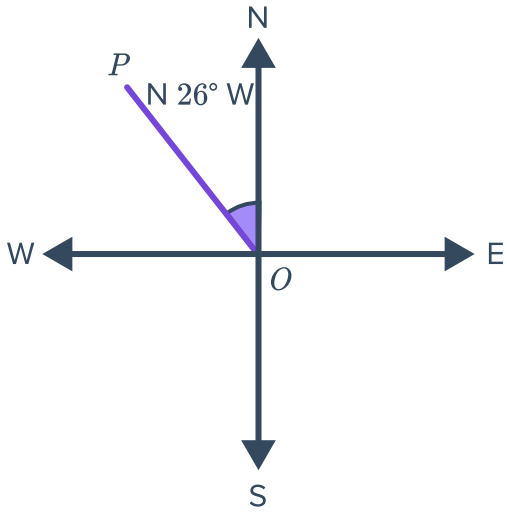
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9 Determine whether the following diagrams show a true bearing of 296° T for the point P from the origin O.

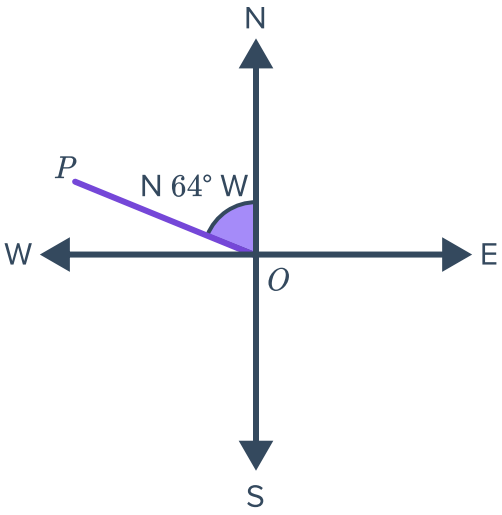
a



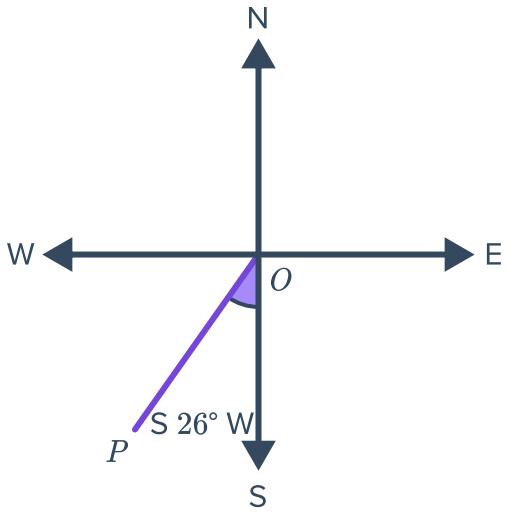
b



c



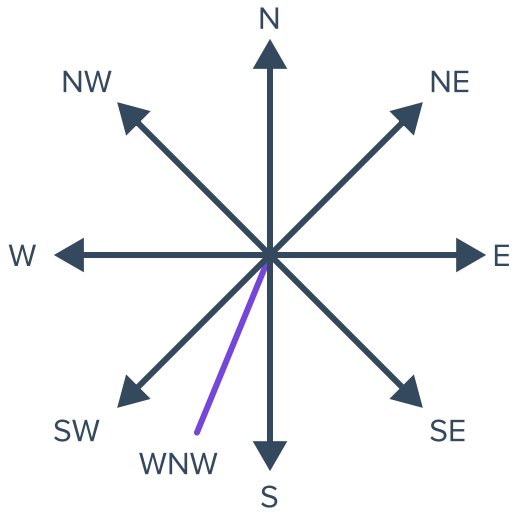
d



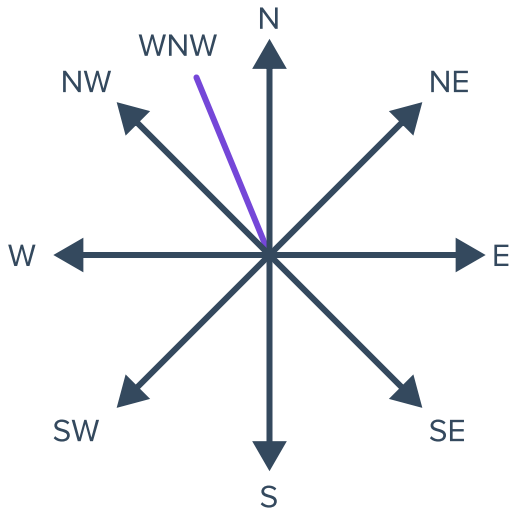
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10 Determine whether the following diagrams display WNW:

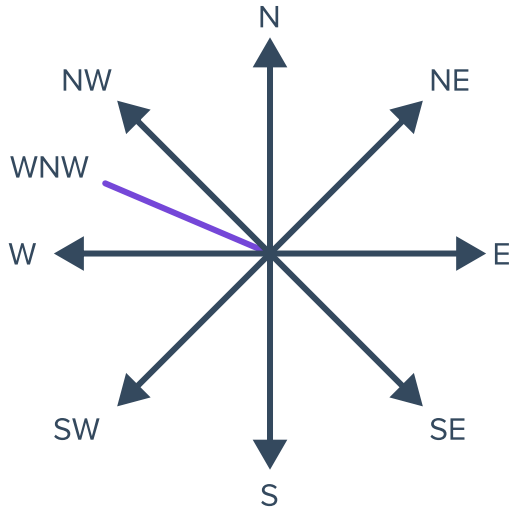
a



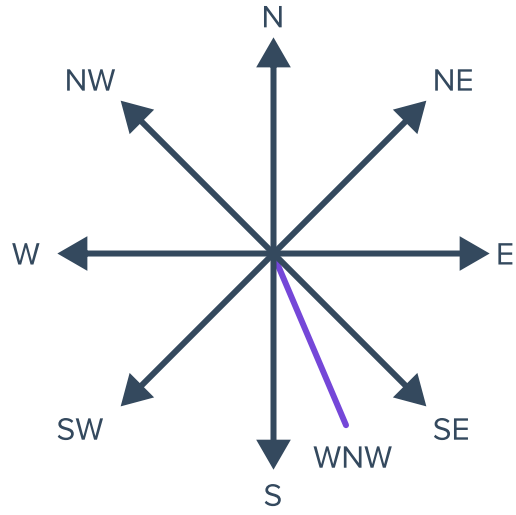
b



c



d



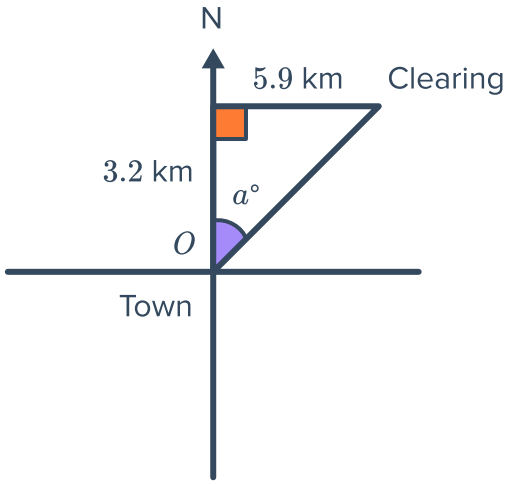
11 If a person walks at a compass bearing of SW, and then turns around to go back in the direction they came, what direction are they heading now?

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Applications

12 Consider the following diagram:

If the bearing of the clearing from the town is a°, find a to the nearest degree.



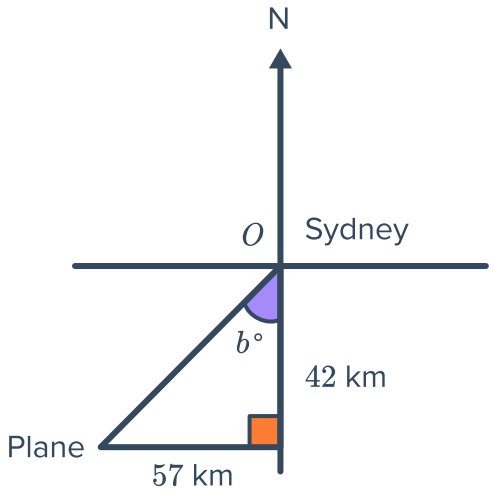
13 An aeroplane sets off on a bearing of 051°T. After some time, it needs to turn back and head to its original position. On what true bearing must it travel?

14 A boat sets off on a bearing of 125°T . After some time, it needs to turn back and head to its original position. On what true bearing must it travel?

15 Shortly after take-off, a plane is $42\ \text{km}$ south and $57\ \text{km}$ west of the airport in Sydney that it left from:

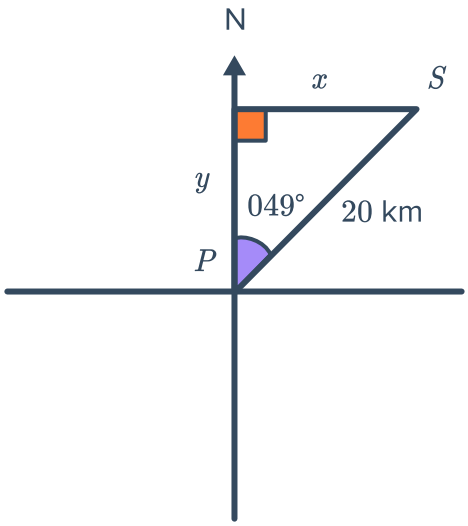
a Find the size of the angle marked b, to one decimal place.

b Find the true bearing of the plane from the airport.



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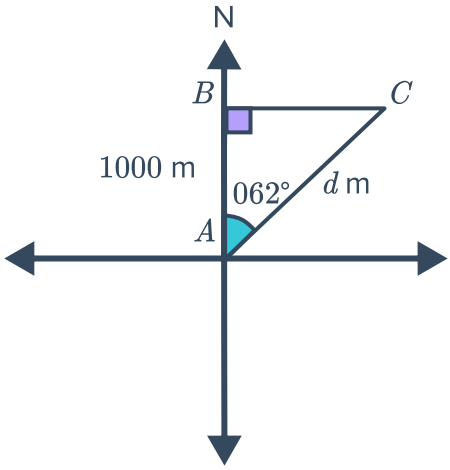
16 The position of a ship S is given to be $20\ \text{km}$ from P, on a true bearing of $049^\circ\ \text{T}$. The position of the ship can also be given by its (x, y) coordinates.



a Find the ship's x-coordinate to one decimal place.

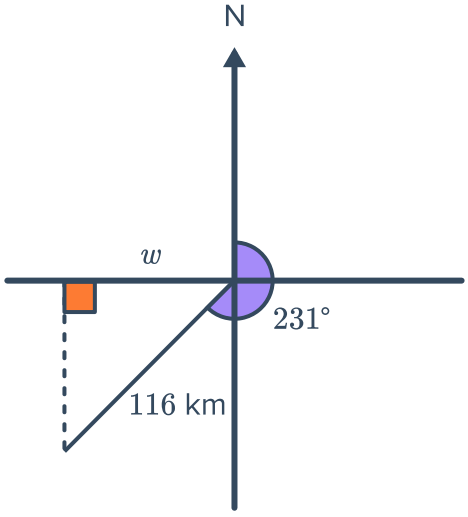
b Find the ship's y-coordinate to one decimal place.

17 On an orienteering course, Beth runs 1000 metres north from point A to point B, then turns east and runs to point C. If the true bearing of C from A is 062°T and the direct distance Beth must run to get back to point A is d metres, find d to the nearest metre.



18 Luke sailed for $116\ \text{km}$ on a bearing of 231°.

If w is how many kilometres west he has sailed from his starting point, find w to one decimal place.



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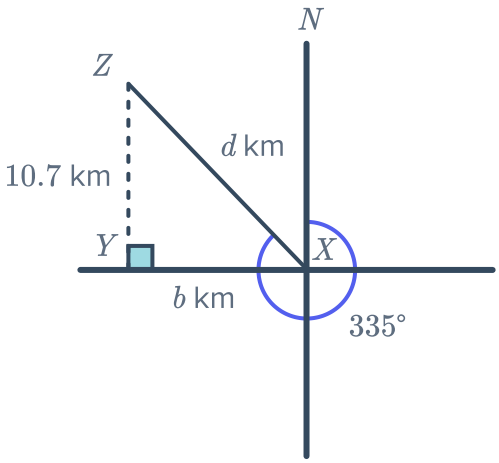
19 During a rescue search, a helicopter flew west from point X to point Y, then changed course and flew 10.7 km north to point Z.

If point Z is on a bearing of 335° T from point X:

a Find the size of $\angle YXZ$.

b If the distance from point Y to point X is b km, find b to one decimal place.

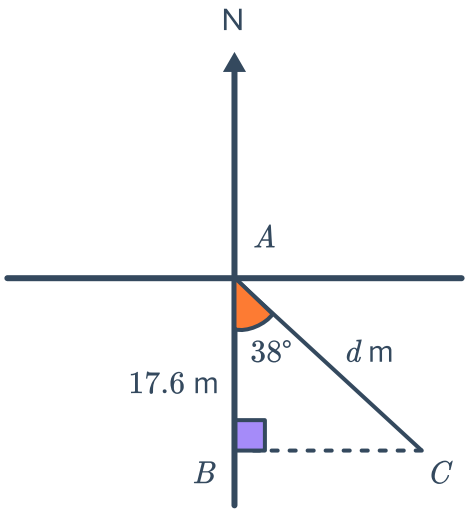
c If the distance that the helicopter must fly between point Z and point X is d km, calculate d to one decimal place.



20 Three television presenters are practising their navigation skills before heading off on an expedition to a remote location.

Belinda at point B is positioned 17.6 m south of Amelia at point A. Carl at point C is due east of Belinda and on a bearing of 142° from Amelia.

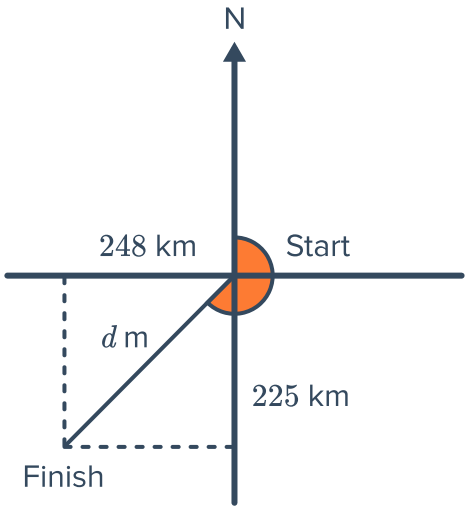
If Amelia and Carl are d m apart, find d to one decimal place.



21 A yacht sailed in a direction so that its final position was 248 km west and 225 km south of its starting point.

a If the true bearing on which the yacht sailed is b°, find the value of b to one decimal place.

b If the boat has sailed a total of d m, find the value of d to one decimal place.



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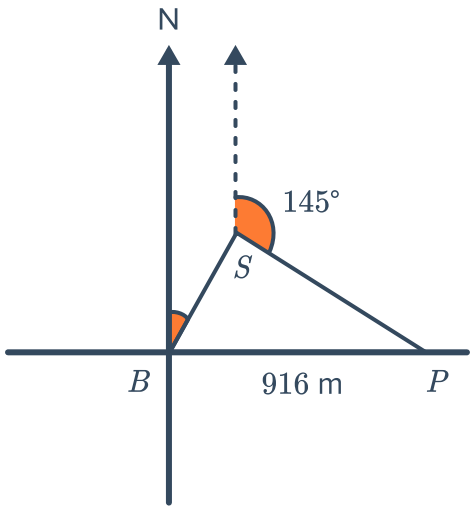
22 In remote locations, photographers must keep track of their position from their base. One morning a photographer sets out from base, represented by point B, to the edge of an ice shelf at point S on a bearing of 055°. She then walked on a bearing of 145° to point P, which is $916\ \text{m}$ due east of base.

a State the size of $\angle BSP$.

b Find the distance BS, to one decimal place.

c Find the distance SP, to one decimal place.

d If the photographer walks directly back to her base from point P, determine the total distance she would have travelled. Round your answer to one decimal place.



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